

Questions			Answers	Marks
8	(a)		After fission, products have smaller nucleons with <u>higher binding energy per nucleon</u> (more stable) , so energy will be released. Accept other equivalent statement.	B1
	(b)		$\Delta mass = 235.123 + 1.009 - (94.945 + 138.955 + 2 \times 1.009) = 0.214u$	B1
			$E = mc^2 = 0.214 \times 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2 = 3.197 \times 10^{-11} \text{J}$	M1
			$E = \frac{3.197 \times 10^{-11}}{10^6 \times 1.6 \times 10^{-19}} = 199.8 \text{ MeV} = 200 \text{ MeV}$	A1
	(c)		- number of reactions = $\frac{6.02 \times 10^{23} \text{ mol}^{-1}}{235 \text{ g mol}^{-1}} \times 1.0 \text{ g} = 2.56 \times 10^{21}$ reactions or - number of reactions = $\frac{1 \times 10^{-3} \text{ kg}}{235.123 \times 1.66 \times 10^{-27} \text{ kg}} = 2.56 \times 10^{21}$ reactions	M1
			$E = 2.56 \times 10^{21} \times 3.197 \times 10^{-11} = 8.19 \times 10^{10} \text{ J}$	A1
	(d)		energy required from nuclear reactions $\frac{500 \times 10^6 \times 60 \times 60}{0.3} = 6.0 \times 10^{12} \text{ J}$	M1
			$\frac{6.0 \times 10^{12} \text{ J}}{8.19 \times 10^{10} \text{ J g}^{-1}} = 73.3 \text{ g}$	A1

